

Erol Akarsu

Senior Software Engineer | .NET, Cloud Applications and Applied AI

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PROFESSIONAL PROFILE

Software architect and engineer with more than 25 years building enterprise applications, secure integrations, data platforms and cloud infrastructure. Professional consulting work includes C#, ASP.NET Core, Microsoft Azure and Azure DevOps, with C#/WCF services backed by SQL Server and Oracle. More recent work spans REST APIs, automated delivery and applied AI at Lowe's, Wipro and Capital One. I stay close to implementation, testing and production support while working with product, architecture and engineering teams. Ph.D. in Computer Science.

SELECTED RESULTS

- Reduced a five-million-product indexing run from 360 minutes to 40 minutes by redesigning the Spark, Couchbase and Solr flow.
- Built secure C#/WCF service routing with password, certificate and token authentication over SQL Server and Oracle.
- Delivered API frameworks, automated regression tests and production pipelines for large commerce and financial-services teams.

TECHNICAL FOUNDATION

- Applications: C#, ASP.NET Core, .NET/WCF, Java, Spring Boot, Python, FastAPI, Node.js, React, Angular and REST APIs.
- Cloud and delivery: Microsoft Azure, Azure DevOps, Google Cloud, AWS, Docker, Kubernetes, Terraform, Packer, Ansible, Jenkins, Git and CI/CD.
- Data: SQL, SQL Server, PostgreSQL, Oracle, MongoDB, Couchbase, Apache Spark and Apache Beam; database-backed services and ETL.
- Applied AI: transformer-based ranking, LLM integration, retrieval-grounded answers, evaluation gates, human review and audit evidence.

PROFESSIONAL EXPERIENCE

Lead Machine Learning Architect and Principal Software Engineer | Lowe's Companies, Inc.

Oct 2020-Present; ML architecture leadership since Jan 2024 | Charlotte / Mooresville, NC

- Lead architecture for transformer-based product-image selection and generative project assistance that answers customer questions.
- Delivered free-delivery, expedited-delivery and promotion features backed by Spark processing, from implementation through release support.
- Improved search freshness through near-real-time Couchbase-to-Solr ingestion; designed workflow applications around Apache Beam and Elasticsearch.
- Plan roadmaps and releases with senior leaders, engineers and designers while balancing customer impact, maintenance and compliance.

Senior Software Architect | Wipro Limited

May 2018-Oct 2020 | Richmond, VA

- Designed Lowe's enterprise search infrastructure on Google Cloud with Terraform, Packer, Ansible and Jenkins.
- Turned a 20-plus-step Spark and Scala indexing flow into a repeatable production pipeline with clusters, load balancers, firewalls and deployment jobs.
- Reduced indexing time for five million products from 360 minutes to 40 minutes using Spark, Couchbase and Solr.

Master Software Engineer | Capital One

Aug 2016-Dec 2017 | Richmond, VA

- Built reusable Spring and Spring Boot service frameworks, API data flows and service-virtualization environments for financial-services engineering.
- Led developer.capitalone.com redesign and regression testing using Node.js, AngularJS, Spring, Protractor and Cucumber.
- Automated API testing from feature specifications through execution with Swagger and graph-based workflow models.

PROFESSIONAL EXPERIENCE, CONTINUED

Senior Software Developer | pExchange

Dec 2017-Mar 2018 | Richmond, VA

- Built a Spring Boot, MongoDB and Angular user-management system with Apache Shiro roles and permissions for program-budget workflows.

Independent Software Consultant | Self-employed

Sep 2003-Aug 2017 | United States

- Used ASP.NET Core, Microsoft Azure and Azure DevOps in professional client engagements.
- Built secure service routing with C#, WCF, SQL Server and Oracle using password, certificate and token authentication.
- Delivered government and commercial web, portal, integration, data-processing and search projects from requirements through release.
- Modernized a government website with Angular and Node.js; delivered Navy portal work on Oracle WCI and WebCenter.
- Reached 99.5% extraction success with XQuery rules applied to 700,000 lines of COBOL across 237 files.

Senior Software Analyst and Technical Lead | Ferguson Enterprises

Jul 2007-Jul 2016 | United States

- Led web-development and enterprise-integration work while remaining active in design, implementation and release support.

Software Engineer | Cisco Systems

Nov 1999-Jul 2007 | United States

- Worked on unified communications software and the enterprise engineering systems that supported its delivery.

SELECTED INDEPENDENT ENGINEERING PROJECTS

Working prototypes and reference applications; these are not presented as customer deployments.

Medicaid Redetermination Operations | Federal-health workflow demonstrator

React, Node.js, PostgreSQL, Docker, REST APIs

- Built case and register workflows for redetermination, eligibility, enrollment, notices, claims and provider payment with role-controlled actions and attributed audit events.
- Added claims adjudication, data exchange and batch SFTP integration to demonstrate end-to-end operational boundaries and evidence handling.

Company Brain | Permission-aware AI knowledge service

React, Node.js / Express, PostgreSQL, OIDC and LLM-provider APIs

- Implemented retrieval over synchronized, versioned sources with connector access controls, tenant isolation and per-query evidence.
- Added grounded-answer contracts, citation allowlists, prompt-injection quarantine, evaluation gates and reviewer approval.

Integrator Control Plane | Secure API and delivery automation

Node.js, React, PostgreSQL, Docker Compose and GitHub Actions

- Built versioned workflows with approved HTTP connectors, encrypted credentials, tenant roles and durable audit history.
- Added idempotent worker runs, bounded retries, dead-letter recovery, migration tests, dependency checks and container builds.

AI Model Cost Orchestrator | Evaluation-led model routing

React, Node.js, Flask, PostgreSQL and Kubernetes

- Designed model routing around catalog and evaluation versions, budget reservations, quality floors and circuit breakers.
- Added approval thresholds, tests and schema migrations to keep model decisions reviewable.

Personal Context MCP | Governed context services

Tenant-scoped context workflow and append-only audit history

- Built explicit version transitions, permission review and optimistic concurrency for shared context sources.
- Protected mutations with idempotency keys and recorded changes in an append-only database audit trail.

SELECTED INDEPENDENT ENGINEERING PROJECTS, CONTINUED

AI Agent Governance Control Tower | Policy and review

Next.js, TypeScript and PostgreSQL

- Built a control plane for agent registration, permission policies, tool-call auditing, human approval, rollback and incident review.
- Kept policy decisions and compliance evidence in durable records rather than relying on model output alone.

MakePDF | Document-processing API

Python, FastAPI, SQLite and isolated OCR workers

- Built verified redaction, bounded streaming intake, isolated OCR jobs, retention controls and signed downloads.
- Added immutable audit records and tests for failure paths in document handling.

AI Code Review and Documentation | Governed developer workflow

React, Node.js, Python, PostgreSQL, Redis and Elasticsearch

- Built governed code-review and documentation workflows with durable records and review points.

NodeGuard | Security evidence and case management

FastAPI, Node.js, React and PostgreSQL

- Combined security telemetry with investigation cases, evidence provenance, permissions and independent review.
- Added external-provider contracts and audit-chain verification.

AI CyberSOC Copilot | Human-reviewed response proposals

Tenant-scoped alert and case workflow

- Drafted evidence summaries and response proposals for administrator review while preserving append-only case history.
- Recorded externally executed results without claiming the application itself performed response actions.

Government Contracts Compliance Platform | Policy-versioned workflow

Independent regulated-operations prototype

- Built obligation checks with authoritative sources, separation of duties and released-decision evidence.
- Preserved decisions with legal holds and verifiable audit exports.

SOLRProject Control Plane | Search deployment safety

Python 3.12, SQLite and Solr

- Staged source changes in SQLite and applied replayable Solr jobs with access-filtered queries.
- Activated verified collection generations through an atomic alias switch.

DELIVERY APPROACH

- Translate requirements into API contracts, data flows, automated tests and deployment steps that teams can operate.
- Use role controls, audit history, evidence review and explicit failure paths in systems handling sensitive workflows.
- Work across engineering, architecture, product and operations; communicate technical tradeoffs clearly to nontechnical stakeholders.

EDUCATION

Ph.D., Computer Science, IDE for Grid Computing | Syracuse University | 1994-1999

M.S., Computer Science | Syracuse University | 1994-1996

M.S., Computer Engineering | Istanbul Technical University | 1991-1993

B.S., Computer Engineering | Ege University | 1986-1991

MACHINE LEARNING COURSEWORK

Neural Networks and Deep Learning; Machine Learning Regression and Classification; Advanced Learning Algorithms; Unsupervised Learning, Recommenders and Reinforcement Learning.